

EEG-Based Multi-Class Motor Imagery Classification Using Variable Sized Filter Bank and Enhanced One Versus One Classifier

Mohammadreza Edalati Sharbaf
Faculty of Biomedical Engineering
Amirkabir University of Technology
Tehran, Iran
edalati@aut.ac.ir

Ali Fallah
Faculty of Biomedical Engineering
Amirkabir University of Technology
Tehran, Iran
afallah@aut.ac.ir

Saeid Rashidi
Faculty of Biomedical Engineering
Department
Islamic Azad University, Science
and Research branch
Tehran, Iran
rashidi.saeid@gmail.com

Abstract— Motor imagery BCI is a system that is very useful to help people with disabilities who can't move their limbs. These systems use brain activity patterns that are made from motor imagery without actual movement. In this paper, we proposed enhanced OVO structure to classify EEG-based multi-class motor imagery signals. Also, variable sized filter bank is proposed to overcome the weakness of fixed sized filter bank that is used several times. SFFS channel selection is one of the efficient methods which is used to obtain the best channels. The results of four-class classification of BCI competition dataset 2a, show that the performance is improved to 0.63 kappa score.

Keywords- BCI; motor imagery; OVO; filter bank; multi-class

I. INTRODUCTION

Brain Computer Interface (BCI) is a computerized system that acquires produced brain's signals during various mental tasks, extracts and classifies proper features and translates them into appropriate control's signals and sends to external devices. BCI is a real help to persons who have motor disabilities [1]. Various BCI applications are described in [2] included controller devices that convert thoughts into commands in video games and Personal Computers (PCs). Motor Imagery based BCI (MI-BCI) is a system which uses imaginations of movement to control external devices. The most commonly used MI-BCIs are based on electroencephalography (EEG) signals, a process of measuring brain's neural activity as voltage fluctuation along the scalp [3].

When a mental activity, such as execution or imagination of hand movement, is produced, changes occur in the corresponding signal power of μ (8-12 Hz) and β (13-30 Hz) rhythms named Event Related Desynchronization (ERD) and Event Related Synchronization (ERS) of these rhythms [1]. For example, when someone intends to move or imagination of movement the right or left hand, a short period of μ and β rhythms ERD produces in the opposite hemisphere of the brain. Afterwards, a μ and β rhythms ERS occurs in the opposite hemisphere of the part of the body imagined moving [4].

In the present work, a hybrid classification system is proposed to improve the four-class classification results. Also,

variable sized filter bank is used to overcome weaknesses of fixed sized filter bank in preparing signals for feature extraction. Channel selection procedures have been paid less attention in four-class classification. In this paper Sequential Floating Forward Selection (SFFS) is used to select proper channels for feature extraction.

In the next section, we briefly review researches working on MI-BCI systems. Section III consists of introduction of datasets that are used in this research. Pre-processing, feature extraction and classification methods are expressed in this section too and the results of them is in section IV. Section V consists of discussion and conclusion of this paper.

II. RELATED WORKS

One of the most effective and widely used algorithms to find optimal spatial filters is Common Spatial Pattern (CSP) that maximizes variance of EEG signals of one class while minimizes variance of EEG signals of the other class. CSP considers only to discriminate spatially patterns while ignores spectral characteristics of EEG signals. Therefore, some extended CSP have been presented to overcome this weakness of CSP.

So far variants CSP methods such as Common Spatio-Spectral Pattern (CSSP) [5], Sub-Band Common Spatial Pattern (SBCSP) [6], Filter Bank Common Spatial Pattern (FBCSP) [7] have been presented. FBCSP, which presented the best performance on the BCI competition IV dataset 2a, extracts CSP features from signals filtered using filter bank that is composed from non-overlapping frequency bands with an equal bandwidth. Later, adaptive FBCSP [8] was proposed to extend FBCSP so that the filter bank be subject-specific.

The works done to improve motor imagery classification results are not only limited to feature extraction procedures, but preprocessing of EEG signals, feature selection and classifier's structure have important role in classification's accuracy. For example, Qiu et al [9] used Sequential Floating Forward Selection (SFFS) for channel selection. Kam et al. [10] proposed

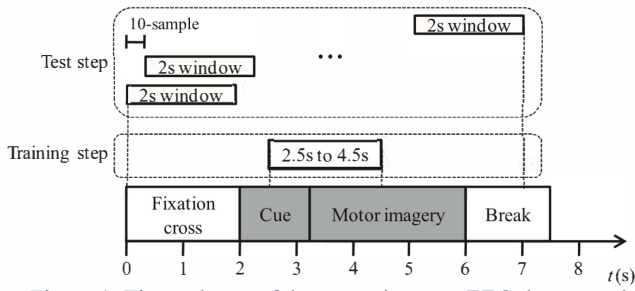


Figure 1. Time scheme of the motor imagery EEG dataset and samples used for training and test [10]

a method to find time-frequency dependent spatial filter that select the best filter by Fisher criterion. Mobaien et al. [11] proposed a new structure to detect class's label by One Versus Rest (OVR) classifiers in multi-class classification.

III. MATERIALS AND METHODS

A. Data Description

We use dataset which consists of EEG data of nine subjects. The cue-based BCI paradigm consists of four different motor imageries as follows: the imagination of movement of the left hand (class 1), right hand (class 2), both feet (class 3) and tongue (class 4). EEG data of each subject recorded in two sessions on different days. Each session comprised of six runs separated by short breaks. Each run consists of 48 trials (12 trials for each class) which will be 288 trials in each session [12]. It means there is 72 trials for each motor imagery in each session.

Timing scheme of paradigm illustrated in Figure 1. These signals are recorded from 22-channel in accordance with the international 10-20 system at 250 Hz sampling rates. They were bandpass filtered between 0.5 Hz and 100 Hz (with the 50 Hz notch filter enabled). The EEG channels of recording signals is shown in Figure 2. This data consists of training and evaluation sections that recorded in two days [12]. As shown in Figure 1, the time segment of 0.5-2.5 s after the onset of the visual cue is used for training. Also, 2 seconds sliding window is used to evaluate the system.

B. Sequential Floating Forward Selection (SFFS)

Multi-channel signals used in BCI systems contain redundant information and cause inconvenient in practical applications. Channel selection can enhance BCI performance by removing redundant channels [9]. SFFS is an intelligent search algorithm and is known as one of the best methods to feature selection [13].

SFFS is suitable method for EEG channel selection [14]. Starting from X_k , SFFS performs the loop of channel selection. X_k is a subset of channels that contains k channels. Y is universal set of channels. The performance (average of two-class classification's accuracy on training data) is shown by $J(X_k)$. The normal SFFS initiates a channel set (X_k) with low number of channels and firstly adds the effective channel to selected channels (X_k) and later eliminates the noisy channel [15]. Since we want to find better channels than universal channels, we change this algorithm slightly so that the initial

channel set contains all channels and the algorithm firstly eliminates the noisy channel and then adds the effective channel. The search process doesn't stop until the $J(X_k)$ is repeated equally in three consecutive iteration. Figure 3 shows the search process of the modified SFFS algorithm.

C. Variable Sized Filter Bank CSP (VSFBCSP)

FBCSP and SBCSP use a fixed sized bandwidth for each frequency band. Thus, they don't consider the other size of frequency bandwidth that this may weakening performance [16]. So, in this section sub-bands with variable sized frequency bandwidth is studied.

We consider 4-40 Hz frequency band and in accordance Figure 4 design several filters with multiple 4 frequency bandwidth by least-square linear-phase FIR¹ filter. These set of bands act as variable sized overlapping sub-band filters and training EEG signals are band pass filtered by these 45 designed filters.

D. Feature Extraction

In this paper, we use both CSP and CSSP for feature extraction.

Common spatial pattern is a mathematical procedure used in signal processing to discriminate two signals to the new signals that have the maximum possible variance difference. The limitation of this procedure is possibility of discrimination only two classes [17]. Assume we have two motor imagery EEG signals in two matrices that are class A and B. The signals have N channels and T samples. The calculation procedure of CSP is as following:

$$R_a^i = \frac{x_a^i x_a^{iT}}{\text{trace}(X_a^i X_a^{iT})} \quad (1)$$

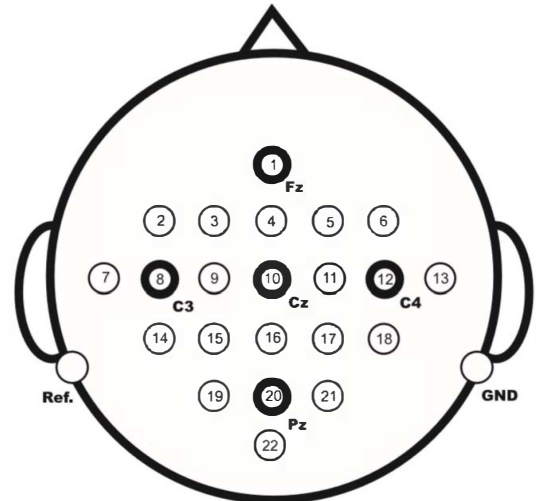


Figure 2. Experimental paradigm of BCI competition IV motor imagery (10-20 system) [12]

¹ Finite Impulse Response

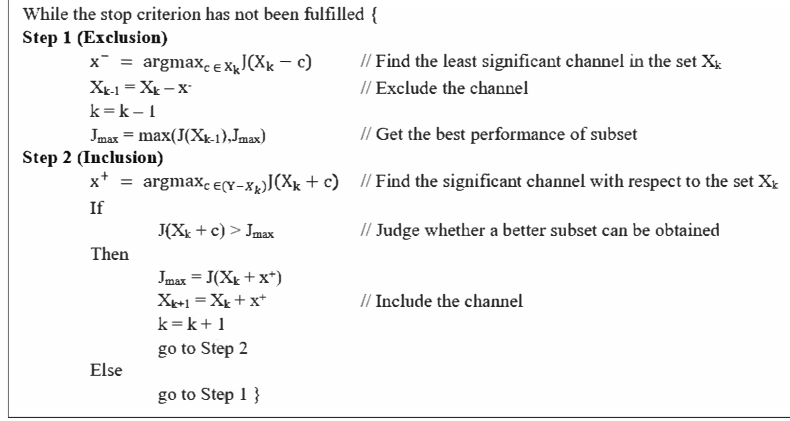


Figure 3. The search process of modified SFFS algorithm

In equation (1), T is transpose operator, X_i is a matrix of trial i of EEG signal and R_a^i is covariance matrix. R_b^i is calculated like equation (1). Afterwards, R_a and R_b is calculated by averaging covariance matrix of trials. So combined covariance matrix is calculated in equation (2):

$$R_c = R_a + R_b \quad (2)$$

Now, we calculate the eigenvalues and eigenvectors of this covariance matrix in equation (3):

$$R_c = U_c \lambda U_c^T \quad (3)$$

U_c is a $N \times N$ matrix that $U_c U_c^T = I_{N \times N}$. λ is a diagonal matrix that diagonal arrays are eigenvalues. Whitening matrix is calculated in equation (4):

$$P = \lambda^{-0.5} U_c^T \quad (4)$$

The whitening matrix causes eigenvalues of transformed matrix be equal one. So now we can calculate transformed covariance matrix in equation (5) and (6):

$$S_a = P R_a P^T \quad (5)$$

$$S_b = P R_b P^T \quad (6)$$

S_a and S_b have same eigenvectors so that $S_a + S_b = P R_c P^T = I_{N \times N}$. thus, the decomposition of S_a and S_b is as following:

$$S_a = B \psi_a B^T \quad (7)$$

$$S_b = B \psi_b B^T \quad (8)$$

Therefore, projected EEG signals is achieving that are proper to discriminate two classes. V is the transforming matrix calculated in equation (9):

$$V = B^T P \quad (9)$$

Only m rows of top and bottom of the matrix V is chosen to generate the mapping matrix W . So the new trial is mapping by equation (10):

$$Z = W^T X \quad (10)$$

Afterwards, feature vector is extracted from the mapped matrix by equation (11):

$$f_p = \log_2 \left(\frac{\text{VAR}_p}{\sum_{p=1}^{2m} \text{VAR}_p} \right) \quad (11)$$

In equation (11), $2m$ is the number of rows of W and VAR_p is variance of each row.

The main idea of CSSP is embed a FIR filter in a spatial filter so that delayed signals ($\delta^T X, \delta^{2T} X, \dots, \delta^{nT} X$) is known as new channels [18]:

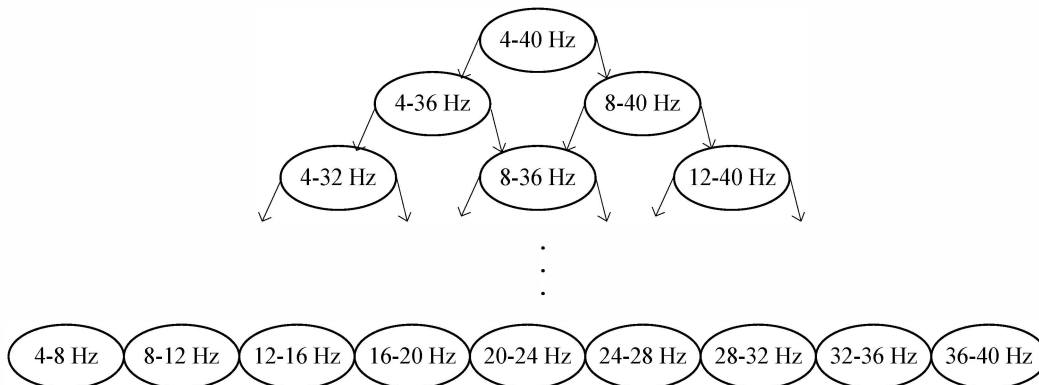


Figure 4. Generated variable sized sub-bands

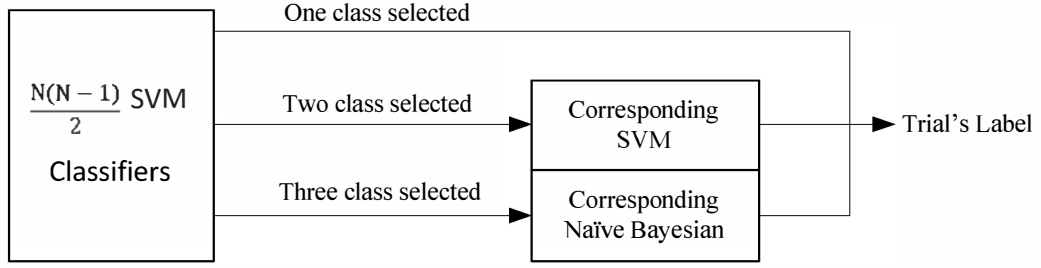


Figure 5. The proposed classifier block diagram

$$\hat{X} = \begin{bmatrix} X \\ \delta^\tau X \\ \vdots \\ \delta^{n\tau} X \end{bmatrix} \quad (12)$$

Where τ is the delay constant, n is the FIR filters order and δ is delay operator. By solving the new generalized eigenvalue, several pairs of $\hat{w}$ corresponding to the largest and smallest eigenvalues can be obtained. Project vector is obtained like CSP algorithm and with that, the signal is projected:

$$\hat{Z} = \hat{w}^0 X + \hat{w}^\tau \delta^\tau X + \dots + \hat{w}^{n\tau} \delta^{n\tau} X \quad (13)$$

Then, the features are extracted according (11) equation.

It is worth noting that with insufficient number of features, CSP and CSSP algorithms tend to overfit, i.e. they learn the noise in training set rather than signal [18]. This matter shows the importance of the channel selection before using CSP and CSSP algorithms.

E. Feature Selection

The Mutual Information-based Best Individual Feature (MIBIF) algorithm is based on filter approach [7, 19]. The mutual information of each feature with its label is computed and sorted in descending order. Then, the k first features are selected. The important note is that because extracted features are pairs, each feature is selected by MIBIF algorithm, its pair must be selected too. For more details refer to [7].

When features are selected from training set, the selected frequency sub-bands are specified too. Thus, only selected frequency sub-bands filters are used to filter test set. This makes evaluation of test set be faster.

F. Classification

One Versus One (OVO) or pairwise classifiers have been used frequently for multi-class motor imagery classification [20, 21]. This classifiers use $\frac{N(N-1)}{2}$ classifiers for N -class classification that each classifier discriminates one class from the other class. The procedure of this system is that each class wins the classification result more than others is the trial's label. The main weakness of this algorithm is unfolded when more than one class are specified as the trial's label. It means that more than one class have the most and equally win in comparison to other classes. In four-class classification, it is happening when two or three classes win two classifiers' result.

The hybrid classification system is proposed to overcome this weakness. If two classes win more than others, the classifier that discriminate these two classes with their trained classifier, specifies the trial's label. But if three classes win equally, the existence classifiers can't specify the label of trial. Thus, a Naïve Bayesian classifier which is designed by combined OVR extracted features specifies the trial's label.

Since our work is four-class classification, we need to design four Naïve Bayesian and six linear Support Vector Machine (SVM) to specify classes' labels of the trials without any ambiguity.

Figure 5. Shows the block diagram of the proposed classifier system. This system can overcome disability of the traditional OVO structure to specify some trial's label.

IV. RESULT

Now, by using mentioned methods the BCI competition IV dataset 2a signals are classified and the results will be compared to other studies.

Table 1. Kappa score of the implemented methods

	Method 1 ^a	Method 2 ^b	Method 3 ^c	Method 4 ^d	Method 5 ^e
Subject 1	0.79	0.80	0.77	0.77	0.77
Subject 2	0.34	0.35	0.39	0.36	0.36
Subject 3	0.78	0.78	0.82	0.85	0.81
Subject 4	0.49	0.51	0.55	0.58	0.54
Subject 5	0.37	0.39	0.37	0.38	0.51
Subject 6	0.25	0.29	0.32	0.38	0.41
Subject 7	0.71	0.74	0.74	0.86	0.84
Subject 8	0.69	0.70	0.71	0.75	0.73
Subject 9	0.63	0.64	0.68	0.59	0.69
Average	0.56	0.58	0.59	0.61	0.63

a. FBCSP with traditional OVO classifier

b. FBCSP with proposed OVO classifier

c. FBCSSP with proposed OVO classifier

d. Variable sized FBCSSP with proposed OVO

e. Variable sized FBCSSP with proposed OVO classifier and SFFS channel selection

A. Parameters Selection

We implemented CSP and CSSP with $m = 2$ that means used four spatial filters to project the signals. Also, the FIR filters order $n=1$ is chosen to implement the CSSP. CSSP with different delay constant $\tau=\{8, 18, 30\}$ is implemented and $\tau=18$ is chosen as the best one and mentioned in the .

Another important parameter selection is corresponded to the number of features that is selected by MIBIF. Since extracted features from each frequency sub-band is four dimensionality, the number of extracted features is selected as a multiple of four. We tried $k = \{4, 12, 20\}$ features that have the most mutual information to the labels and $k = 12$ shows the best performance while variable sized filter bank are used as preprocessing. As mentioned in section III-E, since the extracted features are pairs, the number of selected features are between 12 and 24, subject dependently.

B. Implementation of the Introduced Methods

As shown in Figure 2, the test set is divided into 2 s segments and according BCI competition IV the best segment's result is reported. The introduced methods in this paper are added one by one to show the efficacy of each one.

Firstly, we implemented filter bank CSP (FBCSP) with fixed sized frequency bandwidth equals 4 Hz and the traditional OVO structure is used to classify signals. Then, the proposed hybrid OVO structure is replaced the traditional one. Afterwards, the CSSP algorithm is replaced CSP to show the efficacy of embedding an FIR filter into the CSP filter.

To study the performance of variable sized filter bank, it has been replaced the fixed sized filter bank. So, the classification is implemented by variable sized filter bank CSSP with proposed OVO structure to classify the signals. At last, SFFS algorithm is implemented on training set to select the best channels for each subject.

C. Evaluation Criterion

Since we use dataset 2a of BCI competition IV, the kappa score is used to compare our method with the others. The estimation of the kappa score is [26]:

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (14)$$

Where p_0 is the overall agreement on test trials. This parameter is same as the classification's accuracy. p_e is defined as the chance agreement and calculated as:

$$p_e = \sum_{i=1}^M \frac{n_{.i} n_{i.}}{N^2} \quad (15)$$

Where M is the number of classes and N is the total number of trials. $n_{.i}$ and $n_{i.}$ refer to the sum of elements of i th raw and i th column of confusion matrix, respectively. n_{ij} is the element of confusion matrix that indicates how many trials of class i have been classified as class j .

V. DISCUSSION AND CONCLUSION

In this paper, the OVO-based hybrid classifier is proposed. This classifier eliminates disability of traditional OVO and OVR structures to classify some trials that have ambiguity. The idea of hybrid classifier arose from the disability of classification some trials which can have two or three labels. In this cases, the traditional OVO and OVR can't conclude the trial's label and so the trial is classified as unknown label. This matter causes to reduce the classification's accuracy because of disability of classifier. The proposed hybrid classifier specifies mentioned trials' labels and the improvement that is seen by method 2 in comparison to method 1 in is because of them.

The size of frequency bandwidth of filter bank is case that has not considered too much. The non-overlapping and overlapping bands have been studied as FBCSP [7] and SBCSP [6], respectively. As seen in , by using variable sized filter bank, we can use signal's information which is stored in different sized frequency bandwidth and have a better classification's accuracy. Also, it uses the over-lapping bands too. Although variable sized frequency filter bank causes the training session become time consuming, but MIBIF is used to select perfect bands and features, so the speed of test session doesn't change too much in comparison to FBCSP.

The aim of the channel selection in this paper is improve the classification's accuracy. Therefore, the objective function is the classification's accuracy of the training set and the number of selected channels isn't important. Forasmuch as the CSP and CSSP algorithms are so sensitive to the noise, channel selection eliminates the noisy channels. Thus, CSP and CSSP can produce separable features.

Table 2. Proposed method in comparison to past studies

	Cubero [22]	Ang [7]	Wang [23]	Cubero [24]	Kam [10]	Ghaheri [25]	Proposed method ^a
Subject 1	0.76	0.68	0.67	0.75	0.74	0.72	0.77
Subject 2	0.32	0.42	0.49	0.50	0.35	0.43	0.36
Subject 3	0.76	0.75	0.77	0.74	0.76	0.80	0.81
Subject 4	0.47	0.48	0.59	0.40	0.53	0.57	0.54
Subject 5	0.31	0.40	0.52	0.19	0.38	0.31	0.51
Subject 6	0.34	0.27	0.31	0.41	0.31	0.38	0.41
Subject 7	0.59	0.77	0.48	0.78	0.84	0.76	0.84
Subject 8	0.76	0.75	0.75	0.72	0.74	0.75	0.73
Subject 9	0.74	0.61	0.65	0.78	0.74	0.80	0.69
Average	0.56	0.57	0.58	0.59	0.60	0.61	0.63

a. Variable sized FBCSP with proposed OVO classifier and SFFS channel selection

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